

Miniature Watch-Style FM RADIO

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Radio has remained one of the most enduring and essential communication devices, retaining its relevance to this day. Its evolution—from large, bulky units to compact, portable forms—has

been remarkable. This device pushes that evolution further, delivering one of the smallest wearable and portable FM radios to date.

Two innovative versions have been developed: a watch-style wearable FM radio and another compact enough to fit inside an earbud. The watch-style version is built around the IndusBoard Coin V2, which measures just 3cm in diameter. It features a 3cm GC9A01 round LCD display with a touch interface powered by the CST816S touch driver IC, as illustrated in Fig. 1. The required components for assembling the device are listed in the Bill of Materials table.



Fig. 1: Author's prototype for a watch-like wearable FM radio

```

FMRadioSim.ino  Free_Fonts.h
1  #include <TFT_eSPI.h>
2  #include <CST816S.h>
3  #include "Free_Fonts.h"
4  #include <Wire.h>
5  #include <TimeLib.h> // For real-time clock func
6  #include <RDA5807M.h> // RDA5807M FM Radio librar
7
8  // Initialize TFT and Touch
9  TFT_eSPI tft = TFT_eSPI();
10 CST816S touch(6, 7, 10, 11);
11
12 // Initialize RDA5807M
13 RDA5807M radio;
14
15 // Radio state
16 bool isPlaying = false;
17 float currentFrequency = 87.5; // Starting frequ
18 bool onInitialScreen = true;
19 bool isRadioConnected = false; // Flag to track i
20
21 // Button positions and radius
22 const int buttonRadius = 20;
23 const int prevButtonX = 60;
24 const int prevButtonY = 150;

```

Fig. 3: Code snippet for setting the pins for the touch driver

```

// #define ST7789_DRIVER // Full c // #define TFT_RST // Set TFT_RST to -1 if display
// #define ST7789_2_DRIVER // Minima // For ESP32 Dev board (only tested with GC9A01 d:
// #define R61581_DRIVER // The hardware SPI can be mapped to any pins
// #define RM68140_DRIVER
// #define ST7796_DRIVER #define TFT_MOSI 21 // In some display driver boar
// #define SSD1351_DRIVER #define TFT_SCLK 1
// #define SSD1963_480_DRIVER #define TFT_CS 2 // Chip select control pin
// #define SSD1963_800_DRIVER #define TFT_DC 3 // Data Command control pin
// #define SSD1963_800ALT_DRIVER #define TFT_RST 4 // Reset pin (could connect to
// #define ILI9225_DRIVER // #define TFT_BL 5 // LED back-light
// #define ILI9225_DRIVER // #define TOUCH_CS 21 // Chip select pin (T_CS
// #define GC9A01_DRIVER // #define TFT_WR 22 // Write strobe for modifi
// Some displays support SPI reads via // #define TFT_MISO 19
// bi-directional SDA pin and the libr // For the M5Stack module use these #define lines
// To use the SDA line for reading dat // #define TFT_MOSI 23
// #define TFT_SDA_READ // This o // #define TFT_SCLK 18

```

Fig. 2: Configuration for display in User_Setup.h file of the TFT_eSPI library

BILL OF MATERIALS

Component	Description	Quantity
IndusBoard Coin V2	Main controller	1
GC9A01 round LCD display	3cm diameter touch screen	1
CST816S touch driver IC	For touch embedded on the display	1
RDA5807M FM Stereo radio module	To receive FM signals	1
Audio jack/earbud	To listen to the radio	1
3.7V Li-Po battery	Portable power supply	1

The second version is intended to function as a compact FM radio housed directly within an earbud. A prototype of this version will be described in another article soon.

The objective of these designs is to make the FM radio as small and wearable as possible, whether integrated into an earbud or embedded seamlessly behind the IndusBoard. To achieve the latter, the RDA5807M FM stereo radio module has been selected. This article describes the design of the touch display-based compact FM radio. The next article will explore the second version that fits inside an earbud.

Software

To begin, several essential libraries must be installed via the Arduino Library Manager, accessible from the tools menu. These include: `TFT\_eSPI` for managing the display, `CST816S` for touchscreen control, `RDA5807M` for FM radio